

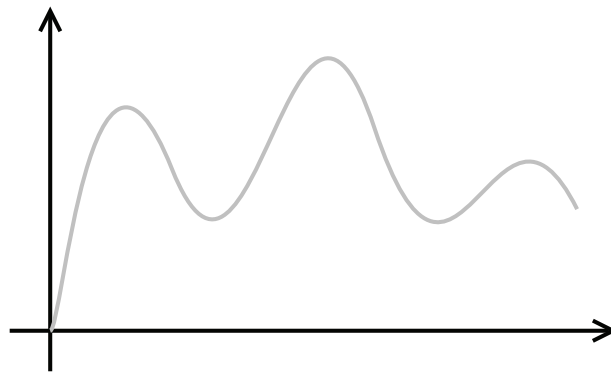
ECE 5200 Syllabus Highlights:

- Course materials at www.carmen.osu
 - syllabus
 - lecture packets
 - * authored by P. Schniter, L. Potter, K. Lee
 - supplementary notes
 - scanned lecture notes
 - assignments and solutions
 - sample test questions
 - links to online resources
- Office hours: TBA (Please respond to a scheduling poll.)
- Text: There exists an equivalence class of good texts
- Classroom goals: provide perspective, context, synthesis, emphases, and structure to make learning efficient
- Email: lee.8763@osu.edu

Introduction:

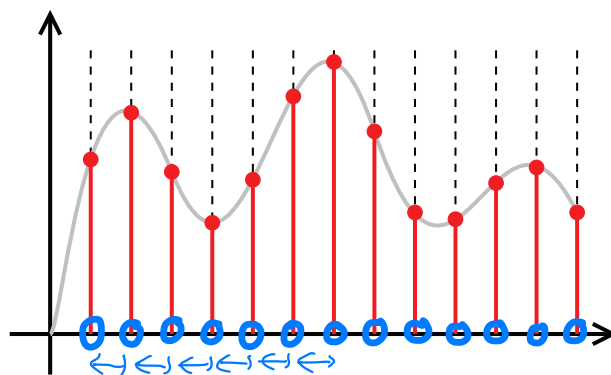
- Much of modern engineering is concerned with signals and systems.
- Roughly, a *signal* is an information-containing waveform (e.g., audio signal, image, digital video stream) and a *system* converts one waveform to another (e.g., low pass filter, mp3 encoder, a simple delay).
- Many real-world signals live in a *continuous* domain (e.g., continuous-time), while much of modern signal processing is done via computation in a *discrete* domain.
- In this course, we develop a solid understanding of signals/systems, especially the relationship between their continuous and discrete representations. The frequency domain will be of particular importance here.
- This course is about understanding/applying *concepts*, not memorizing formulas, and *mathematics* is the language of these concepts. Thus, a major goal of this course is making you comfortable with the mathematics of signals and systems.

Continuous, Discrete-time, and Digital:



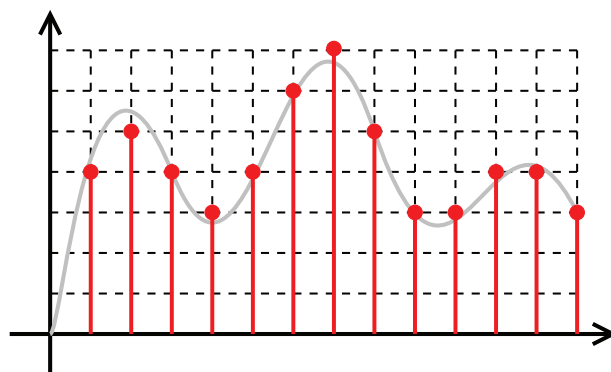
continuous-time signal, $x(t)$

uniform
sampling



uniform grid.
separation between
consecutive elements
is fixed.

sampled signal (discrete-time), $x[n] = x(nT)$

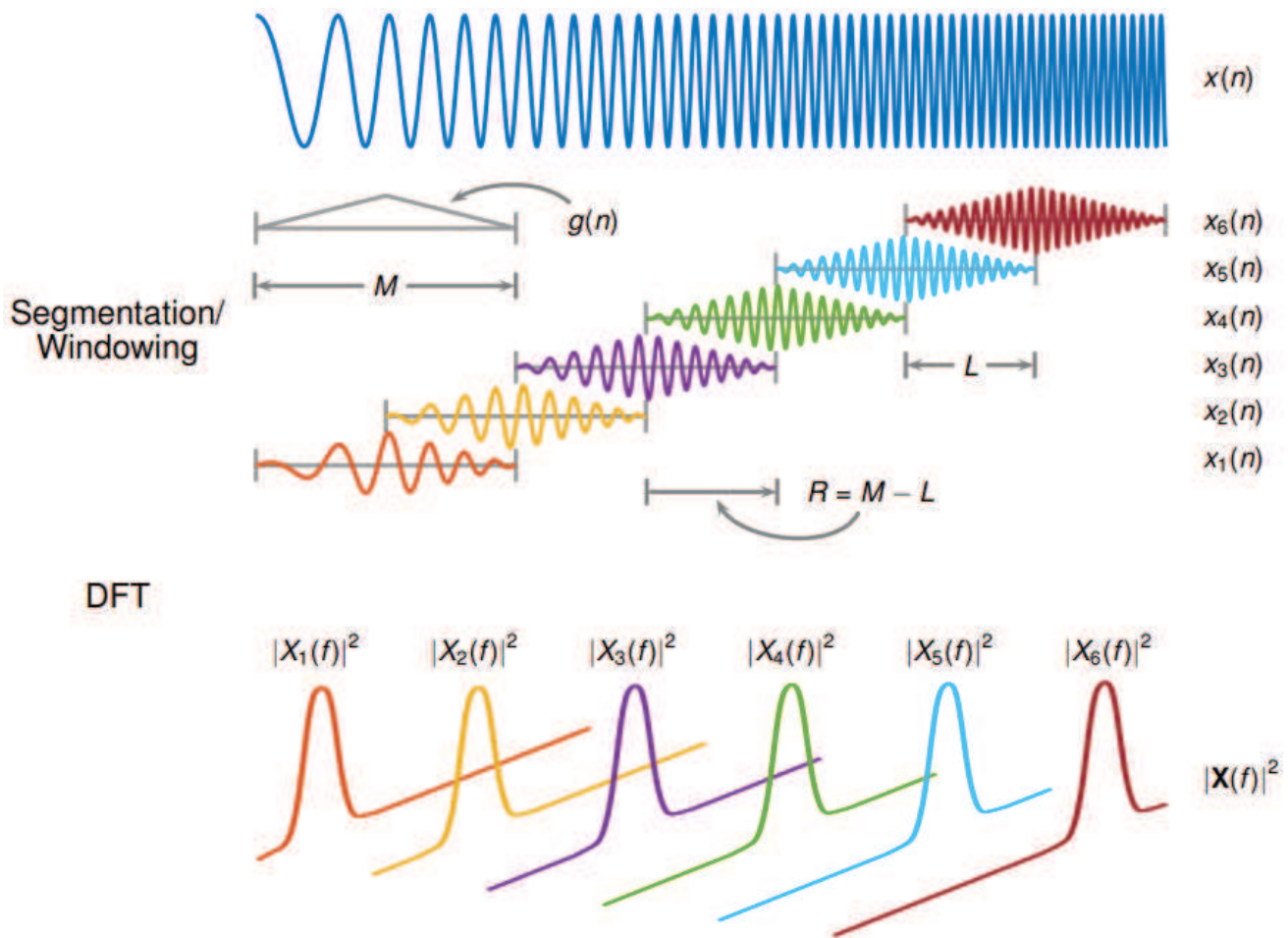


digital signal (discrete-time & quantized)

Outline:

1. Review of Signals/Systems Background
 - (a) Systems: linear, time-invariant, causal, stable
 - (b) Continuous-time transforms: FS, CTFT, Laplace
 - (c) Discrete-time transforms: DTFT, Z-transform
2. Sampling and Reconstruction
 - (a) sampling, aliasing, reconstruction (sinc & practical)
 - (b) upsampling, downsampling, and rate conversion
3. Processing of Finite-Length Signals
 - (a) DFT, circ conv, windowing, spectral analysis
 - (b) matrix/vector formulations
 - (c) FFT, fast convolution, overlap/save
4. Design of Discrete-Time Filters
 - (a) goals: magnitude, group delay, joint
 - (b) FIR designs: window, minimax, least-squares
 - (c) IIR designs: bilinear transform, Prony's method
5. Short-Time Fourier Transform, Filterbanks, Wavelets
6. Sparse Signal Processing and Inverse Problems

Time-Frequency Representations of Signals:

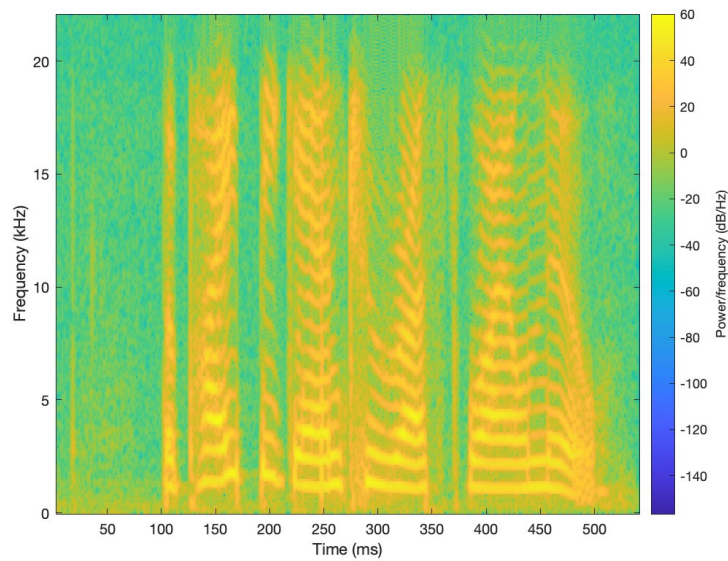


short-time Fourier transform of a chirp signal

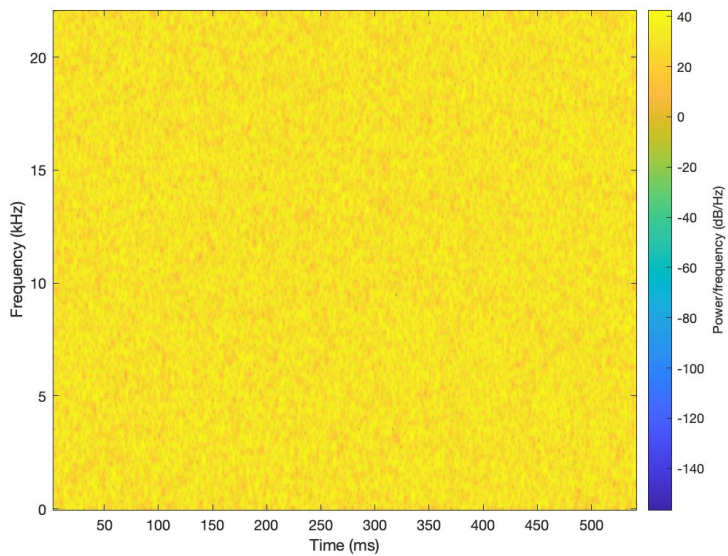
<https://www.mathworks.com/help/signal/ref/stft.html>



voice activity detection
speech recognition

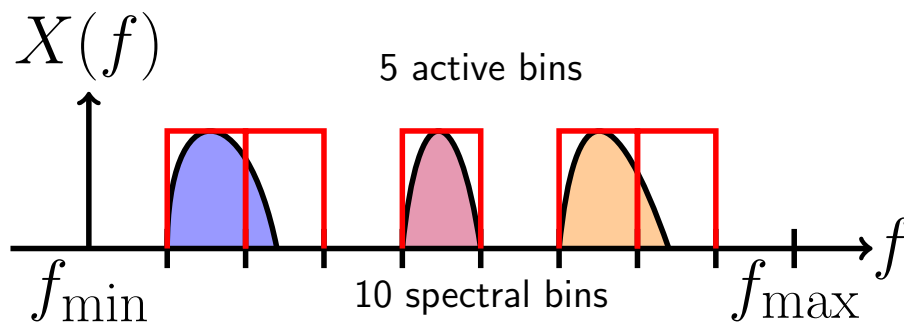


STFT of speech



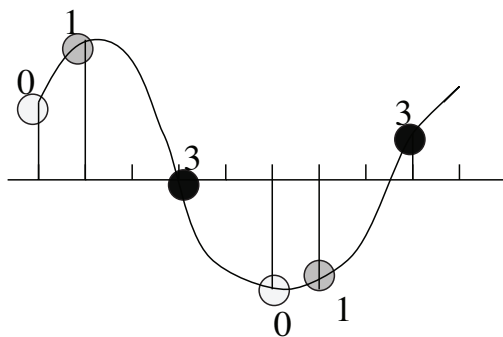
STFT of random noise

Nonuniform Sampling of Multiband Signals:



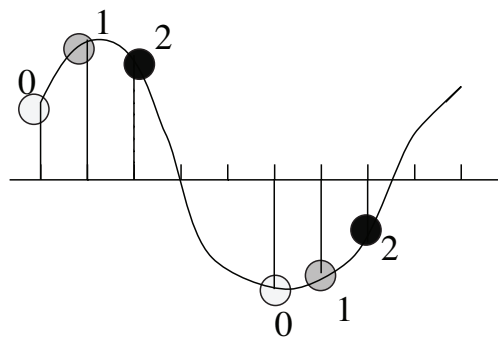
Multicoset sampling

Cosets = 0, 1, 3



Bunched sampling

Cosets = 0, 1, 2



Comments On Homework:

- **Learn by doing.** Learning and retention require repeated access and use of new ideas. Most learning in this course will come from reviewing notes and working through homework exercises. See links at Carmen for study tips.
- Suggested approach:
 1. Review lecture notes. And, as needed, consult textbooks, online resources, and Q&A sessions. Seek multiple explanations and viewpoints.
 2. Attempt to review and understand material *before* attempting the homework exercises.
 3. Make use of Q&A sessions and email for questions.
- *Caution:* Don't short-cut the homework! If you google for answers, or dig through homeworks from previous terms,
 1. You won't learn the material.
 2. You won't do well on the exams.

Complex Numbers:

- What are they?
 - Basically, a complex number is a pair of real numbers
 - Essential to many transforms (Fourier, Laplace, Z)
- Rectangular and Polar notation (say $x \in \mathbb{C}$): *imaginary unit*

$$x = x_r + jx_i$$

$$\text{for } x_r, x_i \in \mathbb{R}, j \triangleq \sqrt{-1}$$

$$= M_x e^{j\theta_x}$$

$$\text{for } M_x \geq 0, \theta_x \in [0, 2\pi)$$

$$= M_x (\cos \theta_x + j \sin \theta_x) \quad \text{via Euler's decomposition}$$

$$= \sqrt{x_r^2 + x_i^2} e^{j \arctan \frac{x_i}{x_r}}$$

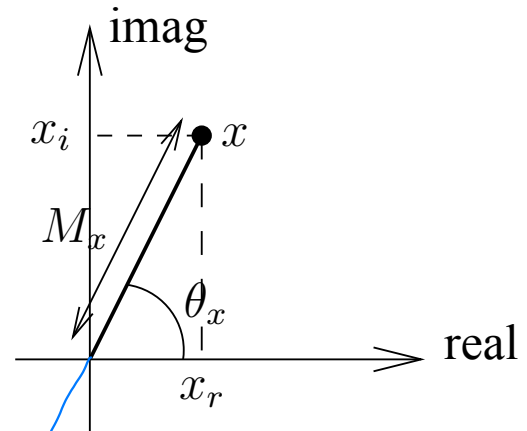


$$\begin{bmatrix} x_r \\ x_i \end{bmatrix}$$

$$r = \sqrt{x_r^2 + x_i^2}$$

$$x_r = r \cos \theta$$

$$x_i = r \sin \theta$$



$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

- Arithmetic: (say $x, y \in \mathbb{C}$)

– Sum:

$$x + y = (x_r + jx_i) + (y_r + jy_i) = (x_r + y_r) + j(x_i + y_i)$$

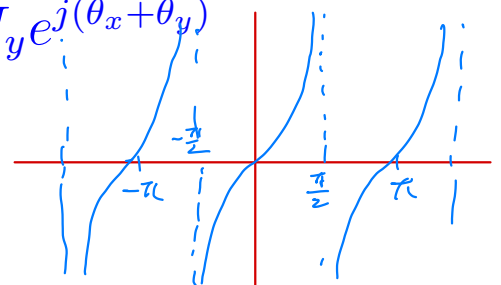
– Product:

$$xy = (M_x e^{j\theta_x})(M_y e^{j\theta_y}) = M_x M_y e^{j(\theta_x + \theta_y)}$$

– Complex conjugate:

$$x^* = x_r - jx_i = M_x e^{-j\theta_x}$$

$$\theta \text{ s.t. } \begin{cases} x_r = r \cos \theta \\ x_i = r \sin \theta \end{cases} \quad \text{"atan2"}$$



Euler's Identity:

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$\Rightarrow \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Why? Use series expansions (Madhava c. 1370; Taylor 1715; Maclaurin c. 1742)

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!} + \frac{z^6}{6!} + \dots$$

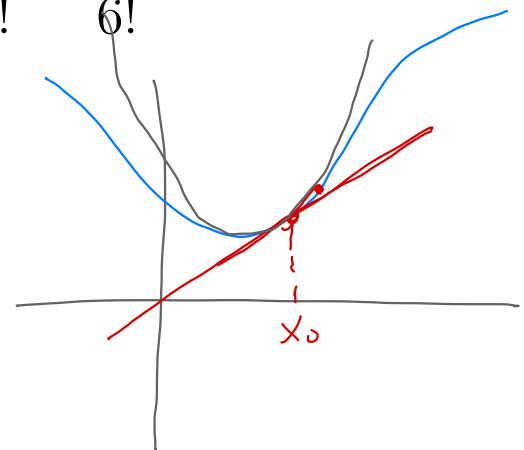
$$e^{j\theta} = 1 + j\theta + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \frac{(j\theta)^5}{5!} + \frac{(j\theta)^6}{6!} + \dots$$

$$= 1 + j\theta - \frac{\theta^2}{2!} - j\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + j\frac{\theta^5}{5!} - \frac{\theta^6}{6!} + \dots$$

$$= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots$$

$$+ j \left\{ \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right\}$$

=



$$f'(x_0) = \lim_{\Delta \rightarrow 0} \frac{f(x_0 + \Delta) - f(x_0)}{\Delta}$$

$$f(x_0) + f'(x_0)(x - x_0)$$

Algebra on real numbers

$\mathbb{R}$: set of real numbers

$$+ : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R} \quad \text{addition}$$
$$(a, b) \longmapsto a + b$$

$$\cdot : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R} \quad \text{multiplication}$$
$$(a, b) \longmapsto a \cdot b$$

binary operator

= maps a pair of real numbers
to a real number

$$a + b = b + a$$

commutativity of $+$

$$a \cdot b = b \cdot a$$

commutativity of $\cdot$

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{distributivity}$$

$$a \cdot b \cdot c = (a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{associativity}$$

Set of column vectors:

$$\mathbb{R}^3 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\}$$

$$\cdot : \mathbb{R} \times \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

- Scalar multiplication

$$\alpha \cdot \mathbf{x} = \alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \alpha \cdot x_1 \\ \alpha \cdot x_2 \\ \alpha \cdot x_3 \end{bmatrix}, \quad \forall \alpha \in \mathbb{R}, \forall \mathbf{x} \in \mathbb{R}^3$$

- Vector addition: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

- Commutativity of vector addition

$$\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

- Associativity of vector addition

$$\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z}, \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$$

- Identity element of vector addition

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \mathbf{x} + \mathbf{0} = \mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

- Inverse elements of vector addition

For each $\mathbf{x} \in \mathbb{R}^3$, $-\mathbf{x} = -1 \cdot \mathbf{x}$ satisfies $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

- Compatibility of scalar multiplication with field multiplication

$$\alpha \cdot (\beta \cdot \mathbf{x}) = (\alpha \cdot \beta) \cdot \mathbf{x}, \quad \forall \alpha, \beta \in \mathbb{R}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

- Identity element of scalar multiplication

$$1 \cdot \mathbf{x} = \mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

- Distributivity of scalar multiplication w.r.t. vector addition

$$\alpha \cdot (\mathbf{x} + \mathbf{y}) = \alpha \cdot \mathbf{x} + \alpha \cdot \mathbf{y}, \quad \forall \alpha \in \mathbb{R}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

- Distributivity of scalar multiplication w.r.t. field addition

$$(\alpha + \beta) \cdot \mathbf{x} = \alpha \cdot \mathbf{x} + \beta \cdot \mathbf{x}, \quad \forall \alpha, \beta \in \mathbb{R}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$